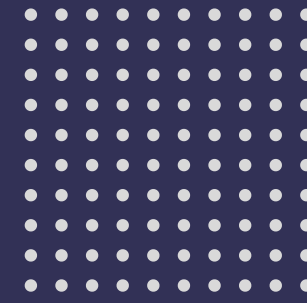
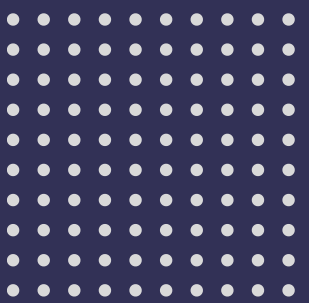


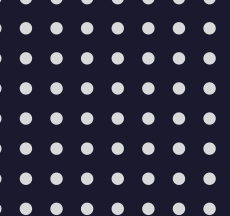


Predicting High Healthcare Expenditure Using MEPS 2022

ALY 6150 — HEALTHCARE ANALYTICS

Group-3
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Introduction & Research Question

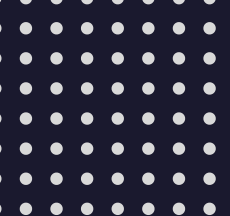
The Problem: In the U.S., a small group of patients drives the majority of healthcare costs. Studies show the top 10% of spenders account for over 60% of total healthcare expenditure.

Research Question:

"What patient-level factors best predict high healthcare expenditure in the U.S. civilian population in 2022?"

Why It Matters:

- Identifying high spenders early enables targeted interventions
- Helps insurers, hospitals, and policymakers allocate resources efficiently
- Critical for reducing waste in the U.S. healthcare system



About the Dataset

Dataset: MEPS HC-243 — 2022 Full Year Consolidated Data File

Source: Agency for Healthcare Research and Quality (AHRQ)

Released: August 2024

Key variable domains:

- Demographics (age, sex, race, region)
- Health status (chronic conditions)
- Insurance coverage
- Healthcare utilization and expenditure

Feature	Detail
Sample size	22,431 individuals
Population	U.S. civilian, non-institutionalized
Coverage	Calendar year 2022
Data collection	Panels 24, 26, 27 — multiple interview rounds

Outcome Variable & Key Definitions

Outcome Variable: high_exp — Binary (0/1)

Value	Definition
1 = High Spender	Total expenditure above median (\$1,605)
0 = Low Spender	Total expenditure at or below median

Key Engineered Variables:

Variable	Definition
multimorbidity	1 if patient has 2+ chronic conditions
female	1 = Female, 0 = Male
older_65	1 if age ≥ 65
college_plus	1 if education ≥ 16 years
Race dummies	6 binary variables from RACEV1X

Chronic conditions tracked:
Hypertension, Coronary Heart Disease, Stroke, Cancer, Arthritis, Asthma

Sample Characteristics & Subgroup Patterns

Table 1: Full Sample (N = 22,431)

Variable	Value
Mean age	42.9 years
Mean total expenditure	\$7,639 (median \$1,605)
Multimorbidity rate	24.70%
Female	52.70%

Table 2: High Spender Rate Across Key Subgroups

Subgroup	High Spender Rate
Female vs Male	55.1% vs 44.4%
Northeast vs South	55.2% vs 45.5%
Multimorbidity vs None	81% vs 40%
Uninsured vs Insured	13.1% vs 52–53%

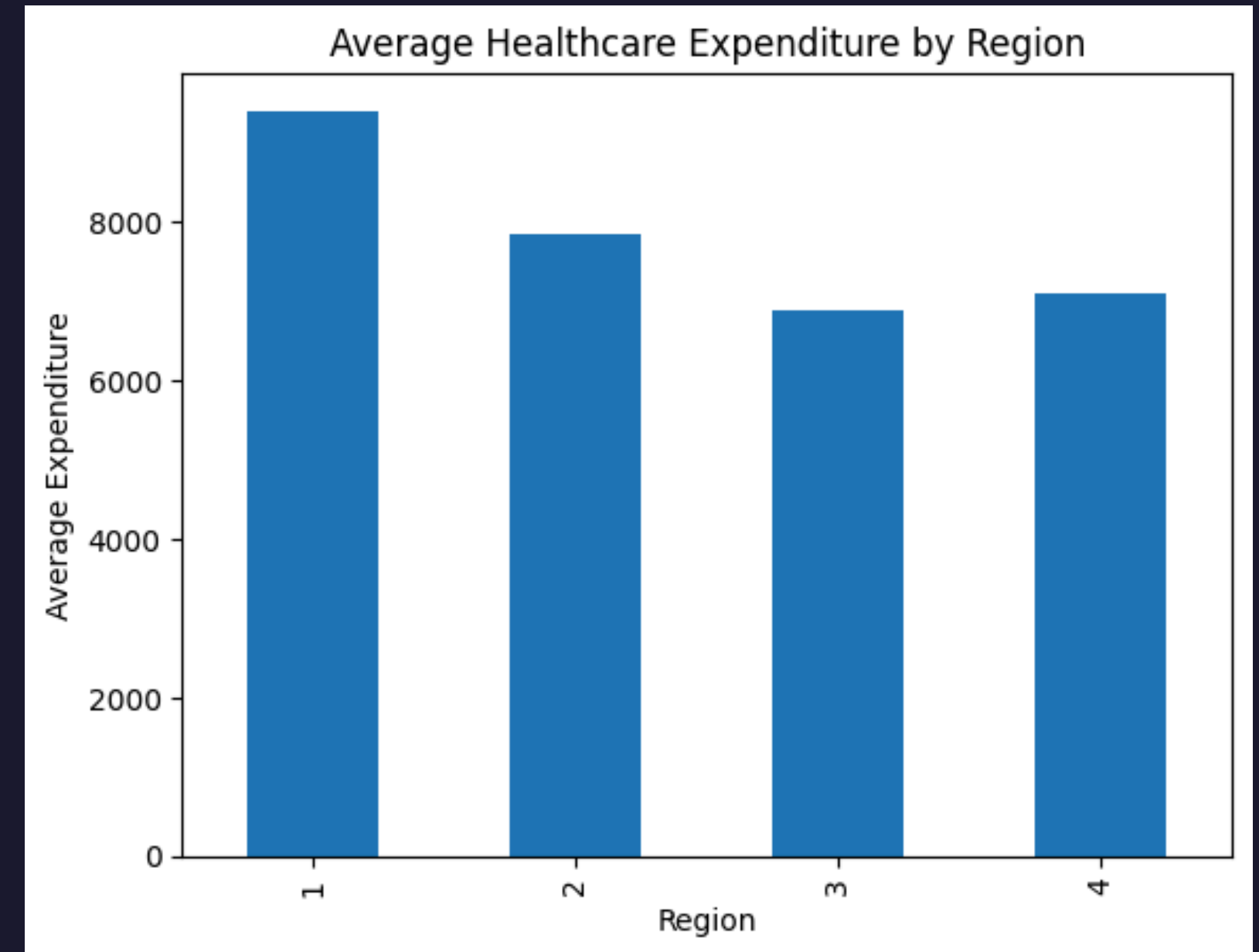
Summary:

- The mean expenditure is nearly 5× the median — confirming extreme right skew driven by a small group of high spenders.
- Females, Northeasterners, and multimorbid patients consistently show higher spending rates across all subgroups.
- Uninsured patients show the lowest spending — reflecting lack of access, not better health.

Average Expenditure by Region

Key Takeaways:

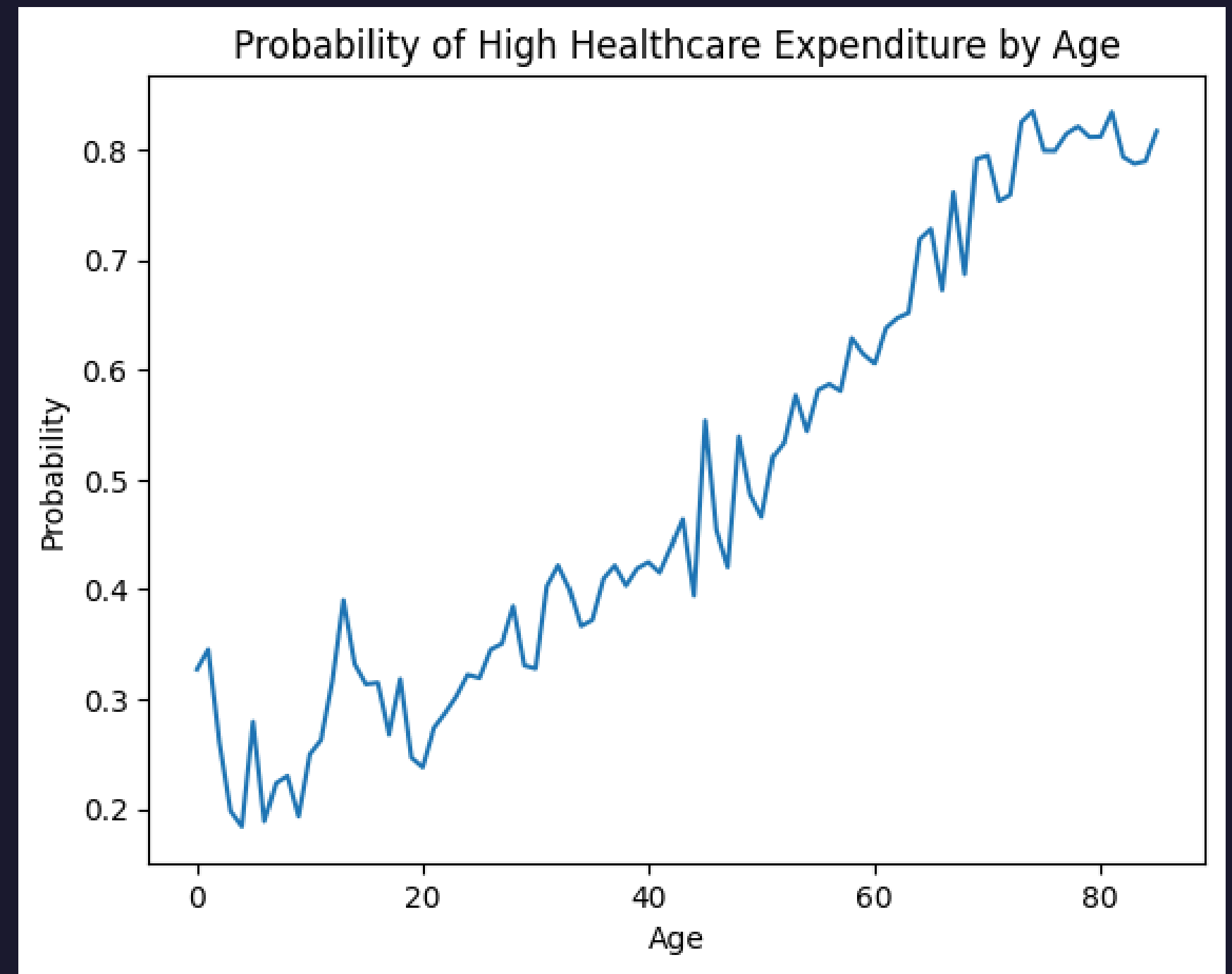
- The Northeast records the highest average expenditure — consistent with higher cost of living and greater specialist care utilization.
- The South records the lowest average spending — despite having comparable multimorbidity rates to other regions.
- Regional variation in spending is not fully explained by chronic disease burden alone — access, cost of care, and provider density all play a role.



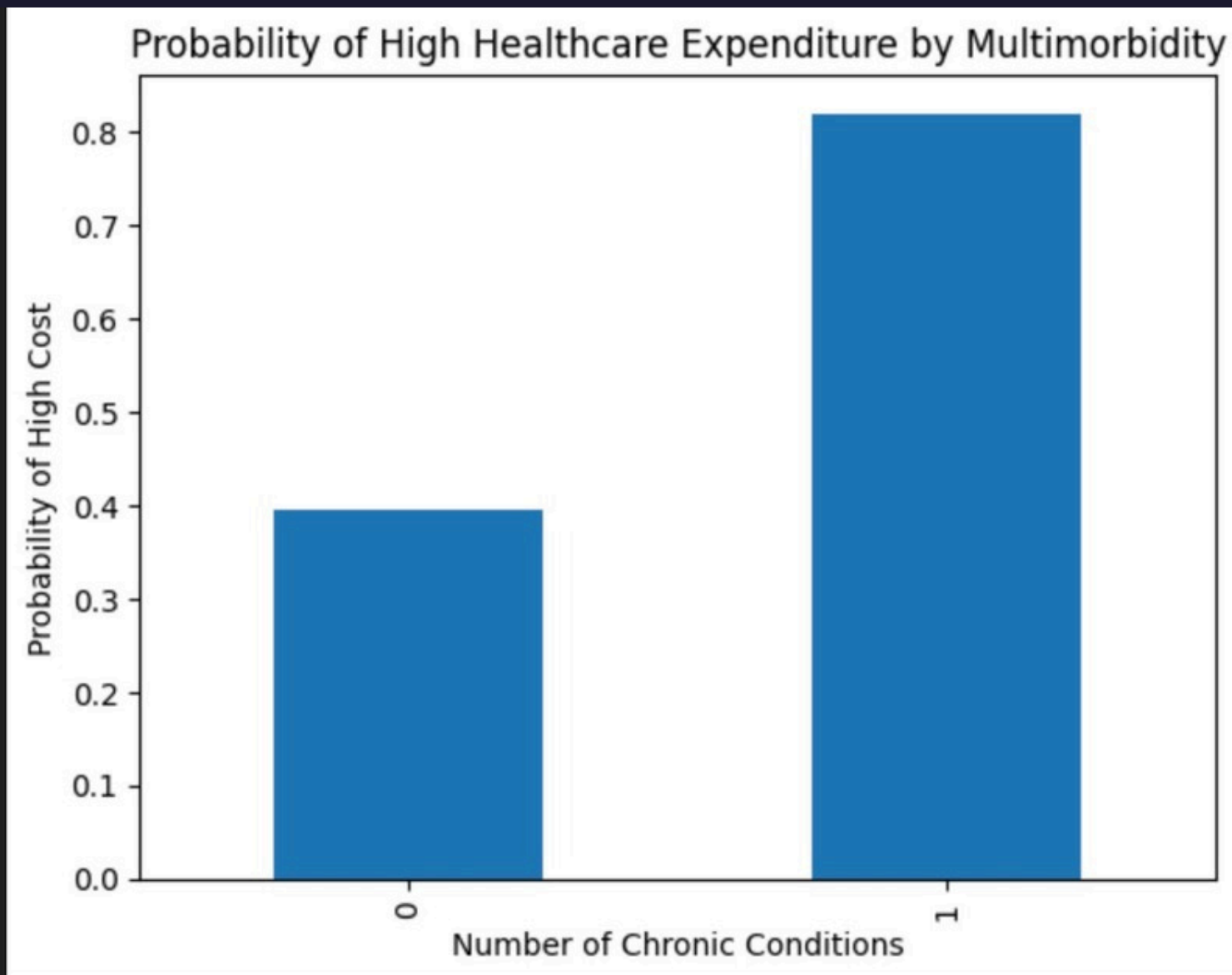
Spending Probability by Age

Key Takeaways:

- A clear upward trend — the probability of being a high spender rises steadily from ~20% in young adults to ~80% at age 80+.
- The steepest increase occurs after age 50 — the critical window for preventive intervention.
- This lifetime trajectory of spending reinforces why age is the second strongest predictor in our models.



Multimorbidity & Geographic Spending

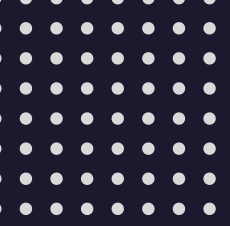


Patients WITH multimorbidity have an 81% high spender rate vs 40% without — a 2× difference and the strongest single signal in the entire dataset.

High Healthcare Expenditure Rate by US Region (MEPS 2022)



- Northeast and Midwest show the darkest shading highest regional spending rates.
- Geographic disparities are real, visible, and consistent with all other analyses.



Hypothesis Testing



Is multimorbidity significantly different between high and low spenders?

	High Spenders	Low Spenders
Multimorbidity rate	Higher	Lower
T-statistic	58.69	—
P-value	< 0.001	—

We reject the null hypothesis. The difference in multimorbidity between high and low spenders is statistically significant ($p < 0.001$). This is not due to random chance.



Logistic Regression

Model: Binary Logistic Regression

Outcome: high_exp (0/1)

Predictors: Age, Sex, Race, Region, Poverty level, Insurance, Multimorbidity

Model fit

Metric	Value	Interpretation
Pseudo R ²	0.13	Predictors explain ~13% of variation
LLR p-value	< 0.001	Model significantly better than guessing
Observations	22,431	Full sample used

Logistic Regression — Odds Ratios

Table: Key Odds Ratios

Predictor	OR	95% CI	Interpretation
Multimorbidity	2.93	2.62-3.27	3× higher odds — strongest predictor
Age (per year)	1.02	1.02-1.03	2% higher odds per year
Female vs Male	1.19	1.08-1.30	19% higher odds
Uninsured vs Private	0.19	0.13-0.30	80% lower odds*
Higher poverty level	0.78	0.67-0.90	Lower spending with poverty

- Multimorbidity is by far the dominant predictor its OR of 2.93 dwarfs every other variable in the model.
- The uninsured OR of 0.19 is misleading it reflects inability to access care, not actual good health.
- Every predictor in the model is statistically significant ($p < 0.05$).

Subgroup Analysis — Male vs Female

Table: Odds Ratios by Gender

Table: Odds Ratios by Gender		
Predictor	Male OR	Female OR
Multimorbidity	2.96	2.91
Age (per year)	1.02	1.02
Uninsured	0.22	0.17
Public insurance	Similar	Similar

- The multimorbidity effect is remarkably consistent across both genders 2.96 for males vs 2.91 for females.
- The uninsured effect is stronger in females (OR = 0.17) women without insurance are even more likely to forgo care than men.
- Gender does not fundamentally change the story multimorbidity remains the dominant driver for both groups.

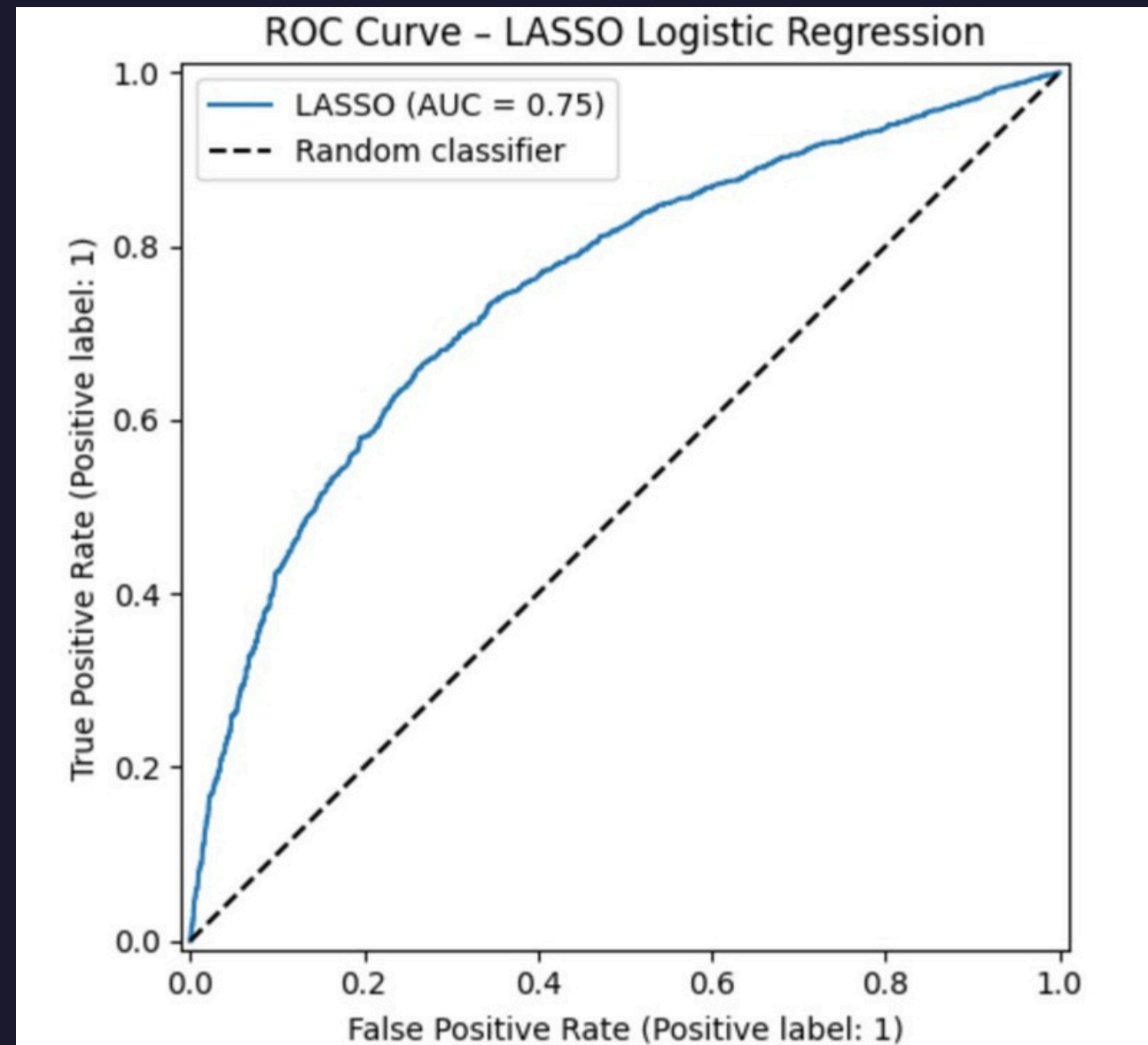
LASSO Model

LASSO Coefficients (standardized):

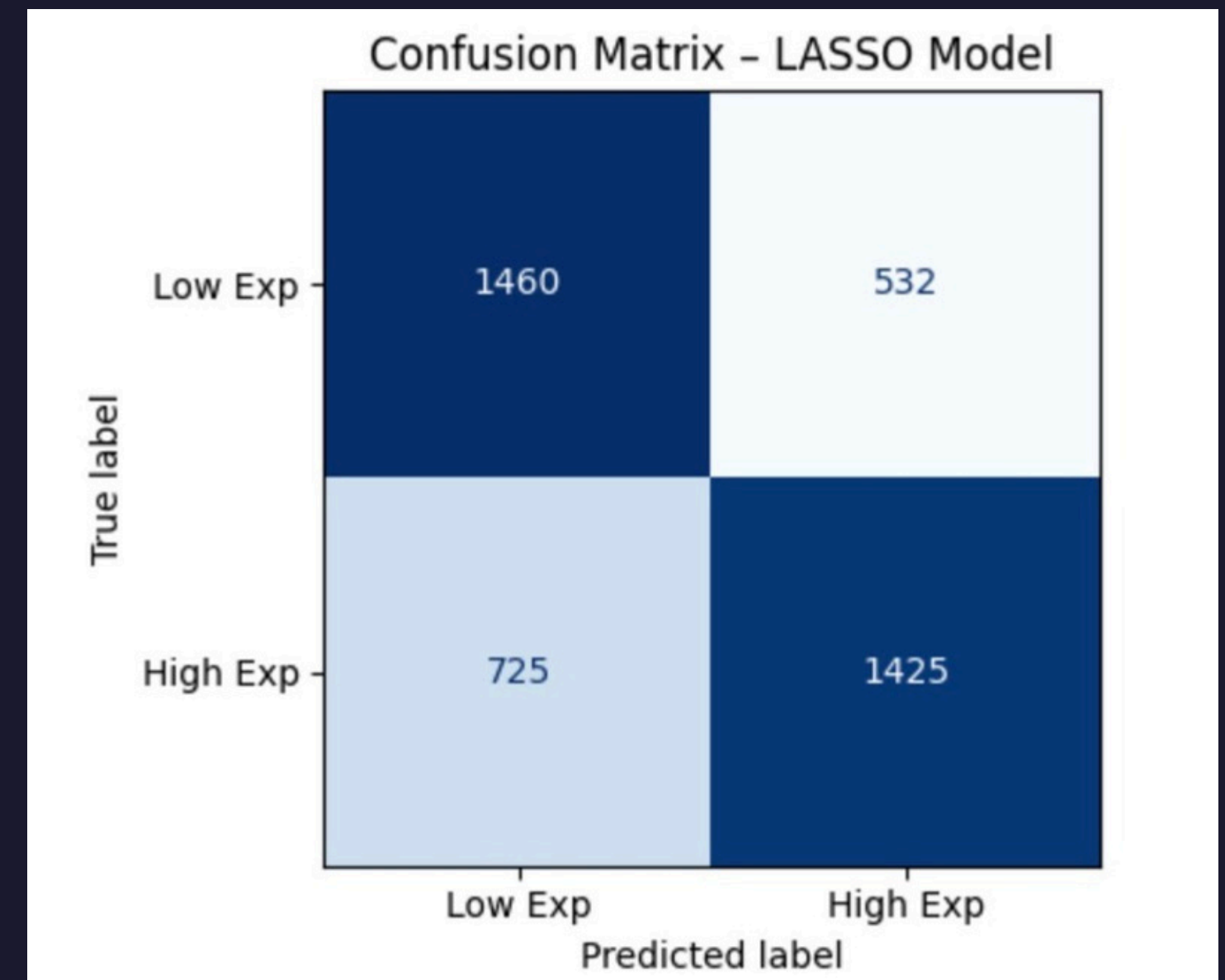
Feature	Coefficient	Rank
Multimorbidity	0.595	 1st
Age	0.430	 2nd
Poverty level	0.190	 3rd
Female	0.175	4th
Education	0.174	5th

- All 5 features survived LASSO regularization confirming they are all genuinely important predictors.
- Multimorbidity ranks first in BOTH logistic regression and LASSO two independent methods, same conclusion.

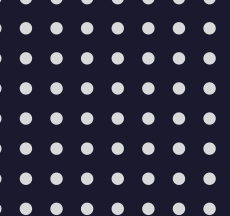
Model Performance



- AUC of 0.76 is a solid result especially with only 5 variables and no utilization data included.
- Adding ER visits, hospitalizations, and prescriptions in future work could push AUC above 0.80.



- 1,460 high spenders correctly identified
- 725 high spenders missed opportunity for improvement



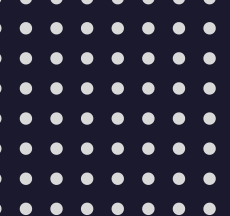
Key Findings & Recommendations

Key Findings:

Finding1 Multimorbidity is the dominant predictor — OR = 2.93, LASSO rank ,#2Age matters — each year adds ~2% higher odds, accumulating significantly ,#3Geographic disparities are real — Northeast spends most, South the least4Uninsured low spending = unmet need, not good health5Gender differences exist but multimorbidity explains most of the gap

Recommendations:

- Target multimorbidity patients for coordinated care management programs
- Prioritize adults 50+ for preventive care — steepest spending increase occurs here
- Expand insurance access — uninsured patients have unmet need, not low need
- Investigate regional cost drivers — especially in the Northeast and Midwest



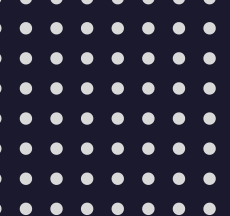
Conclusion

We asked: What factors best predict high healthcare expenditure?

We found: Chronic disease burden — not demographics alone — drives high spending. Multimorbidity (OR = 2.93, LASSO #1) is the single strongest predictor, consistent across every method we used.

Effective cost containment must prioritize chronic disease prevention and management.





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THANK YOU